THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 14

What a Model Is and Why It Works

Exercises

Self-Study Curriculum

1. You have two models of the same system: a 3-parameter model and a 50-parameter model. Both fit the training data equally well. Using MacKay's Bayesian Occam's razor, explain why the 3-parameter model is more strongly supported. Then construct a scenario where the 50-parameter model would actually be preferred.

2. Consider the ideal gas law: PV = nRT. This is a model of gas behavior. Identify: (a) the system, (b) the mathematical structure, (c) the correspondence, (d) what physical features of real gases are absent from the model, (e) when and why the model fails.

3. In machine learning, a model is often evaluated on a "test set" that is drawn from the same distribution as the training set. What *structural* assumption is being made about the relationship between training data and test data? Under what conditions does this assumption fail? Frame your answer in terms of the correspondence between the model's mathematical structure and the system's actual structure.